Franck-Condon Factors Calculation using RKR inversion

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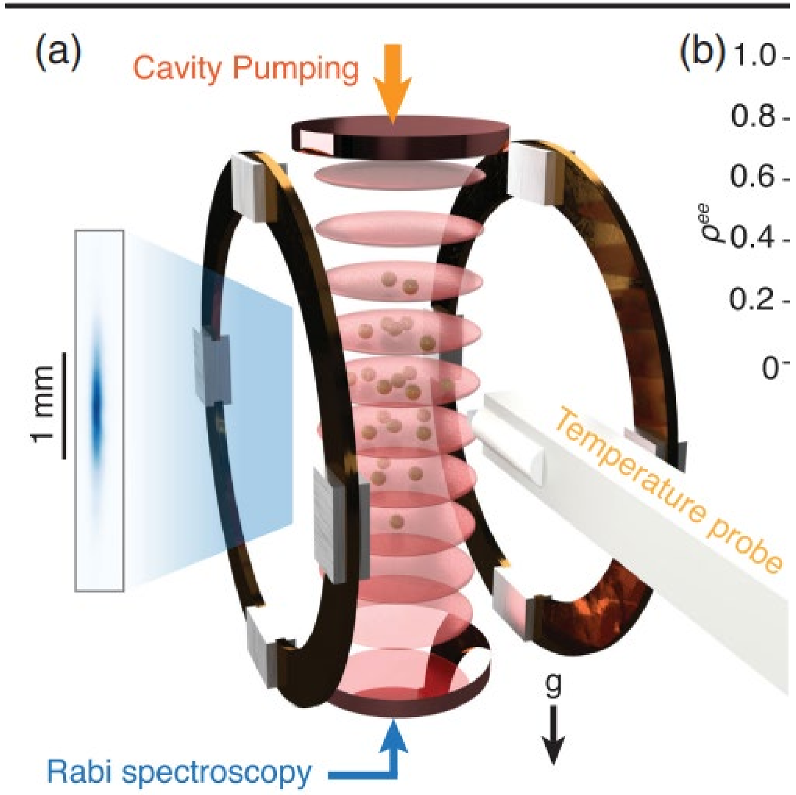
April 2026

PHYS 5070 Final project

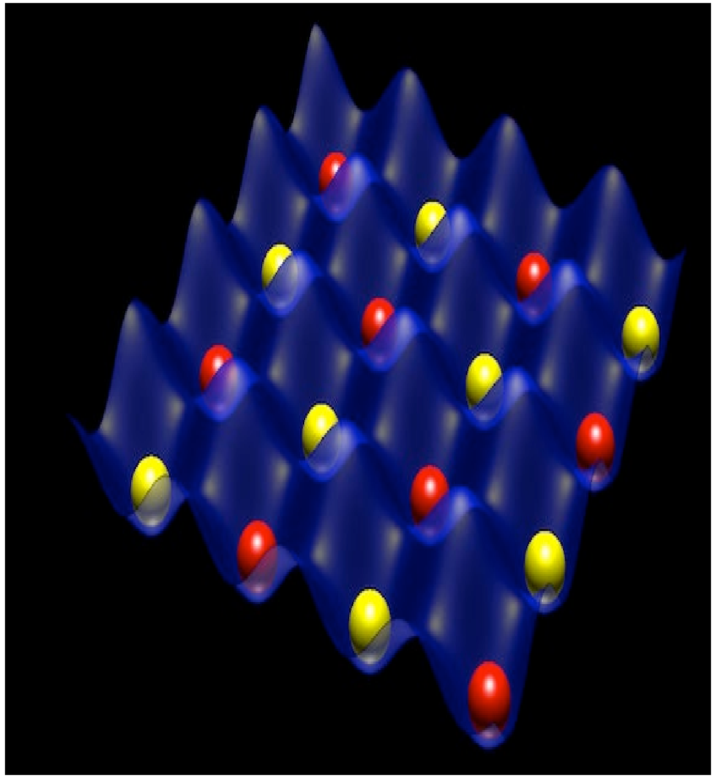
Ultracold Atoms

One of most accurate clocks in the world
1d lattice Strontium atomic clock



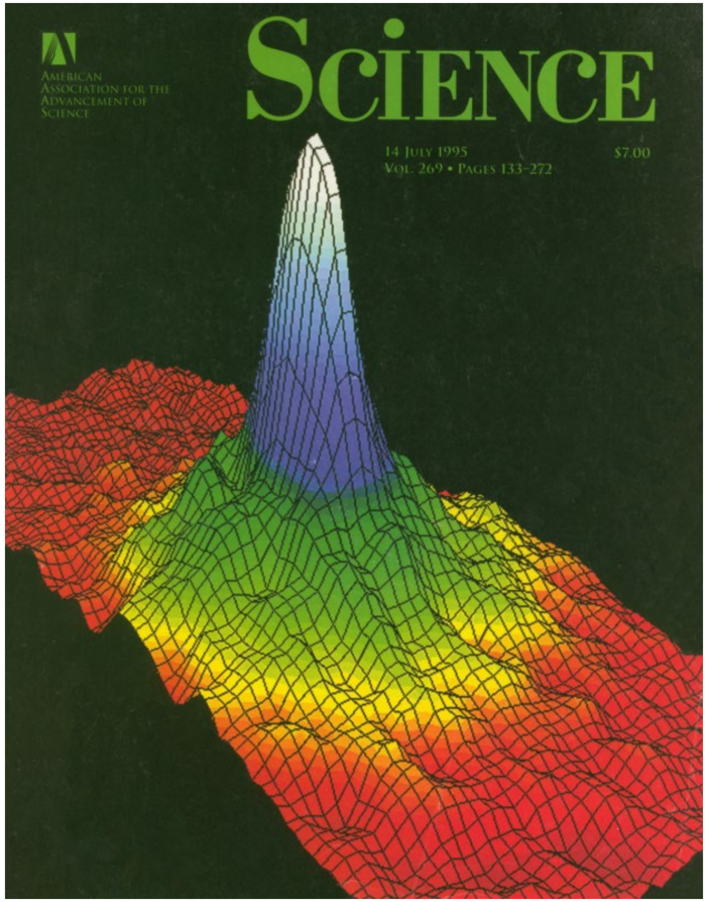
Appeli et al., Phys. Rev. Lett. **133**, 023401 (2024)

Quantum Computation and Simulation with Neutral Atoms



Credit: Nist

First Bose-Einstein Condensate

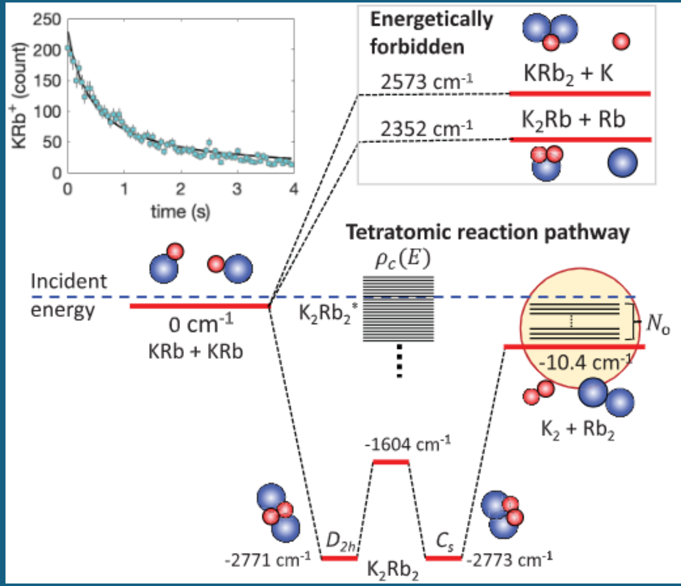


ScienceVolume 269, Issue 5221Jul 1995

Cold Molecules

Cold Chemistry

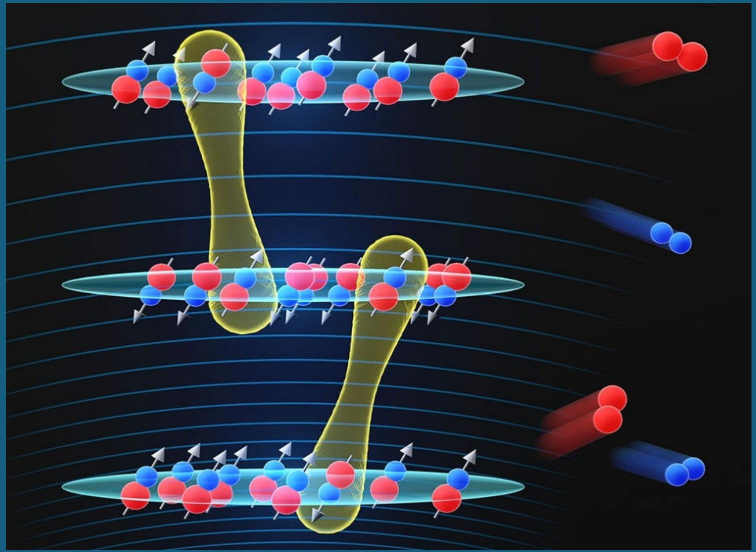
Study collisional effects and complexes through internal and external state control



Hu et al., Science 366, 1111–1115 (2019)

Quantum Simulation

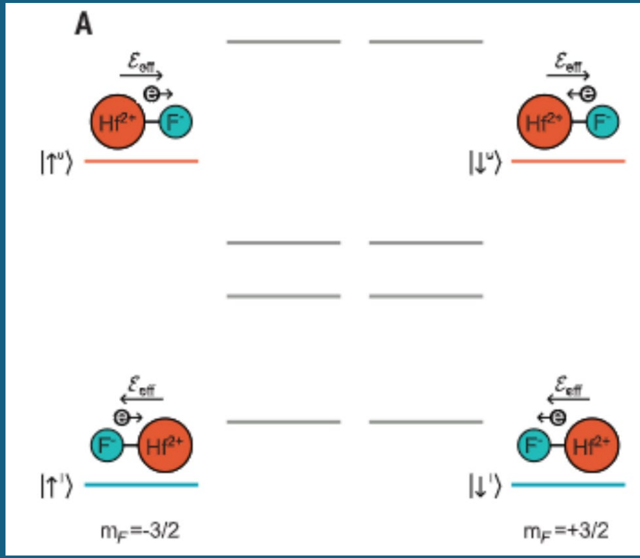
Engineer long-range interactions and explore dynamic phase transitions



The Royal Society:
Quantum Science with Ultracold Molecules (2023)

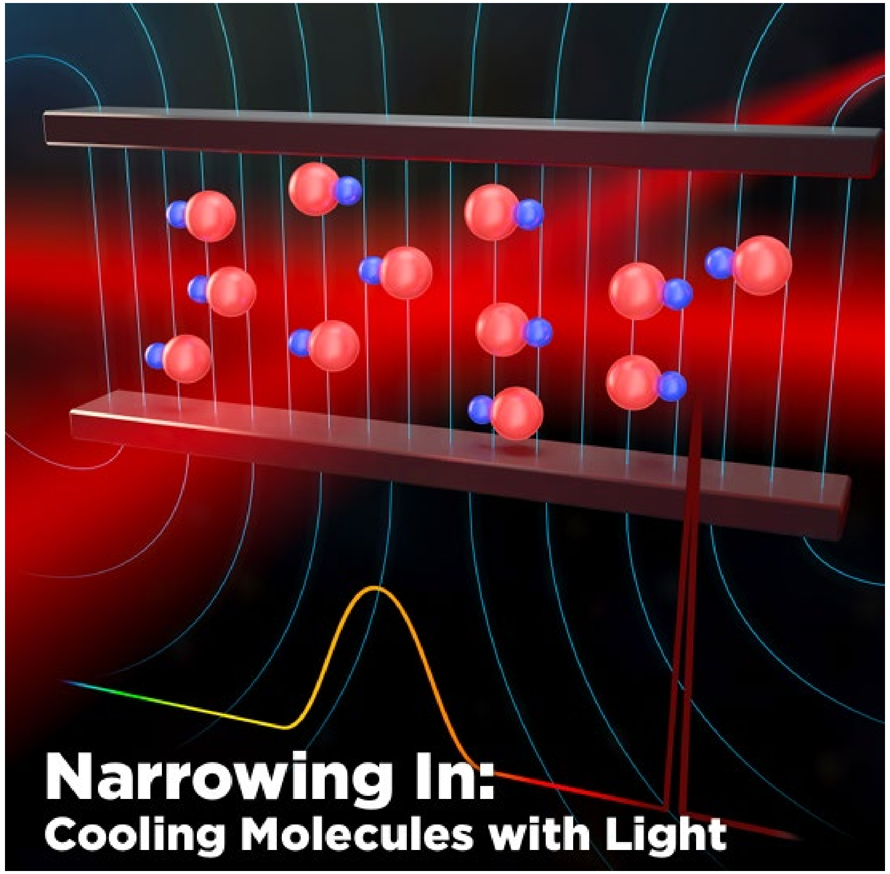
Precision Measurement

Test fundamental physics at record sensitivities on table-top experiments



Roussy et al., Science 381 (2023)

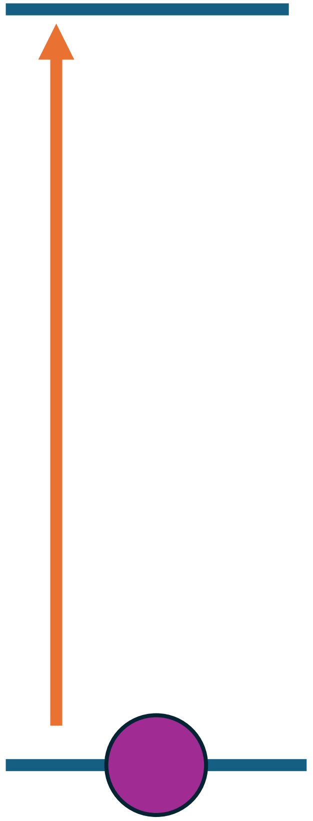
Direct laser-cool of Yttrium Monoxide (YO) molecules

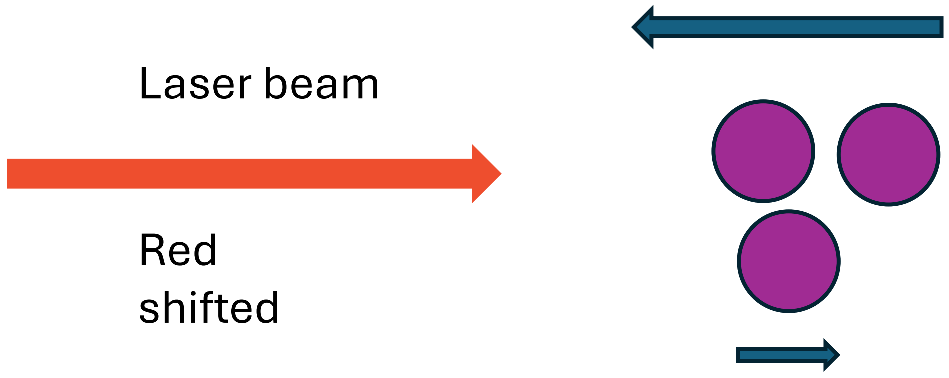


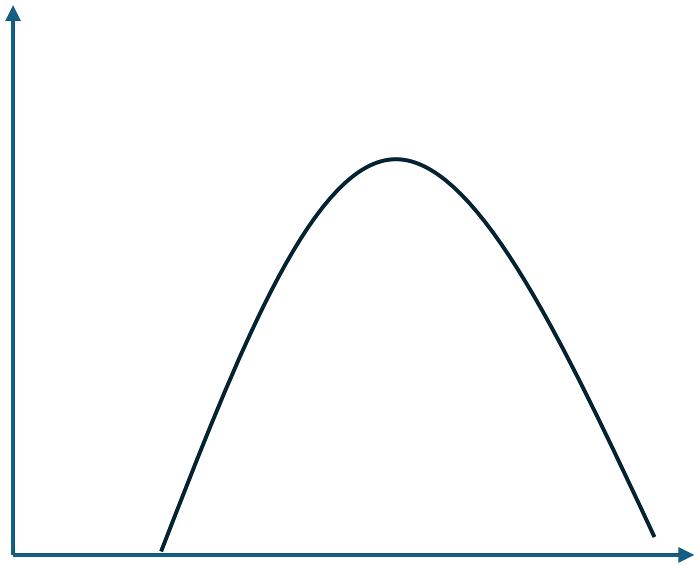
[JILA Light and Matter 2026 Winter]

Ultracold atoms and molecules rely on laser cooling and trapping

Laser Cooling of Atoms







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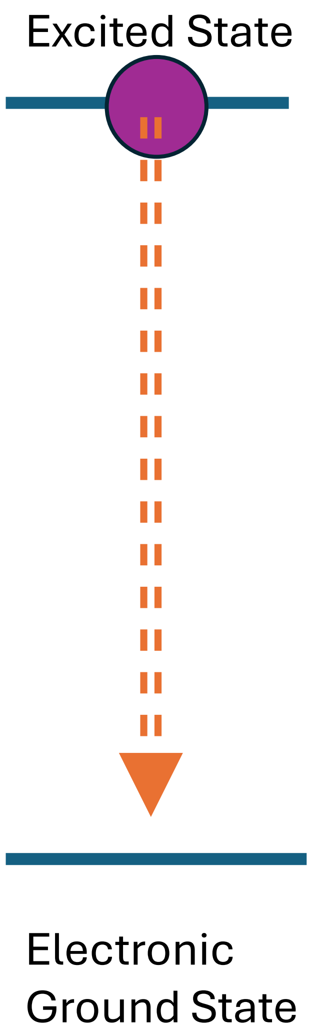

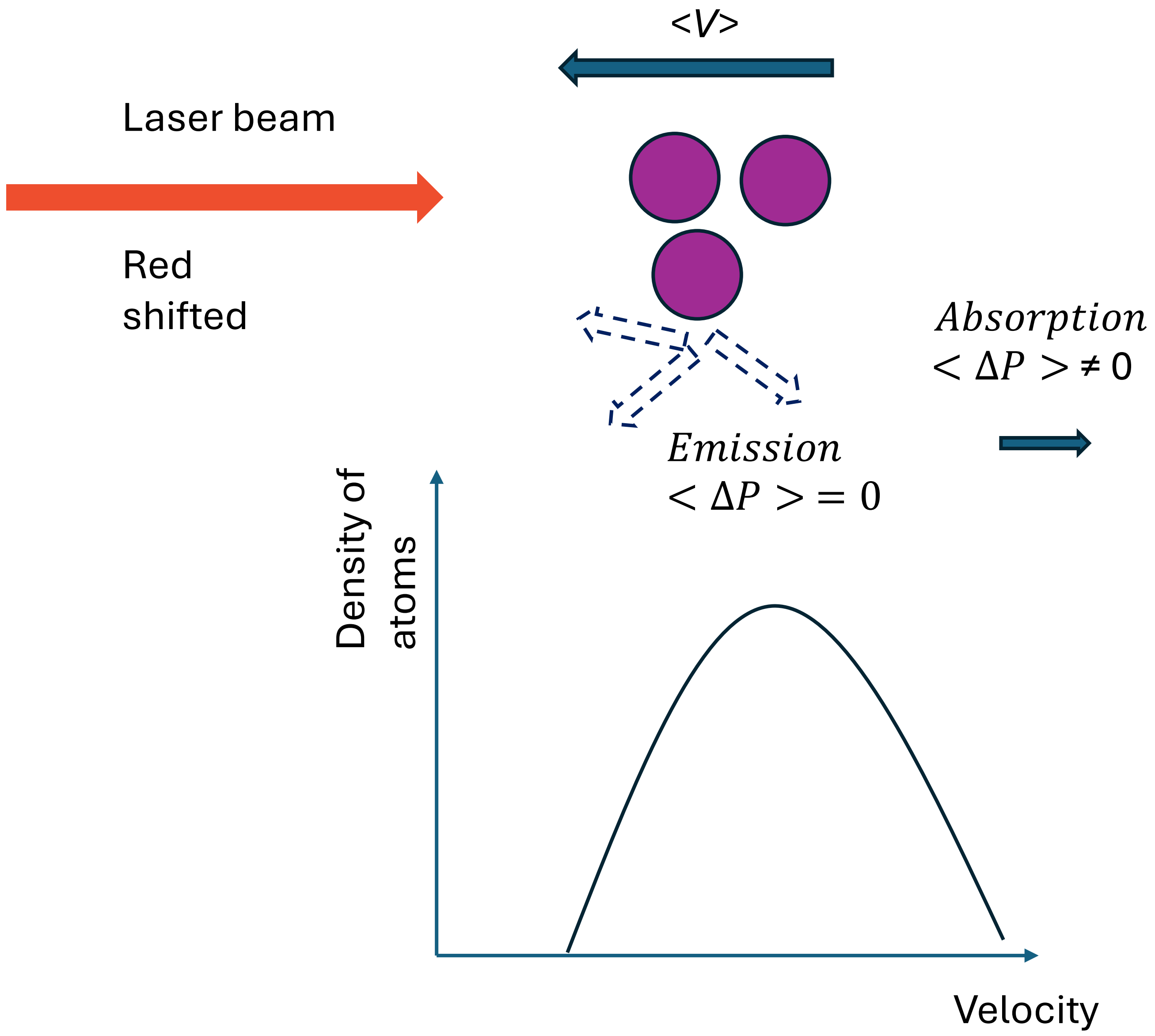

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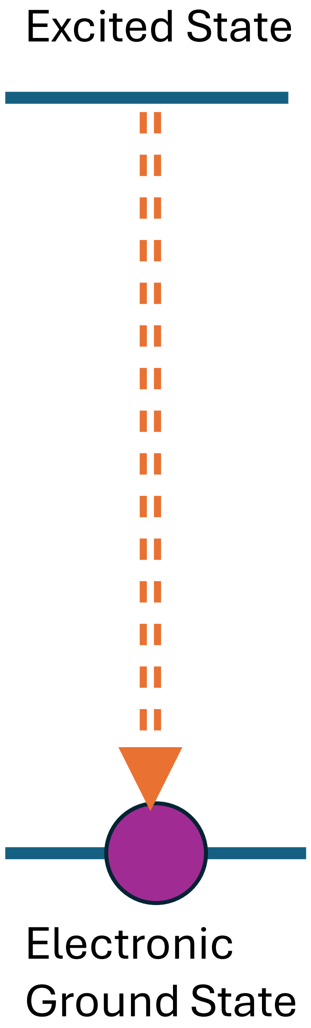

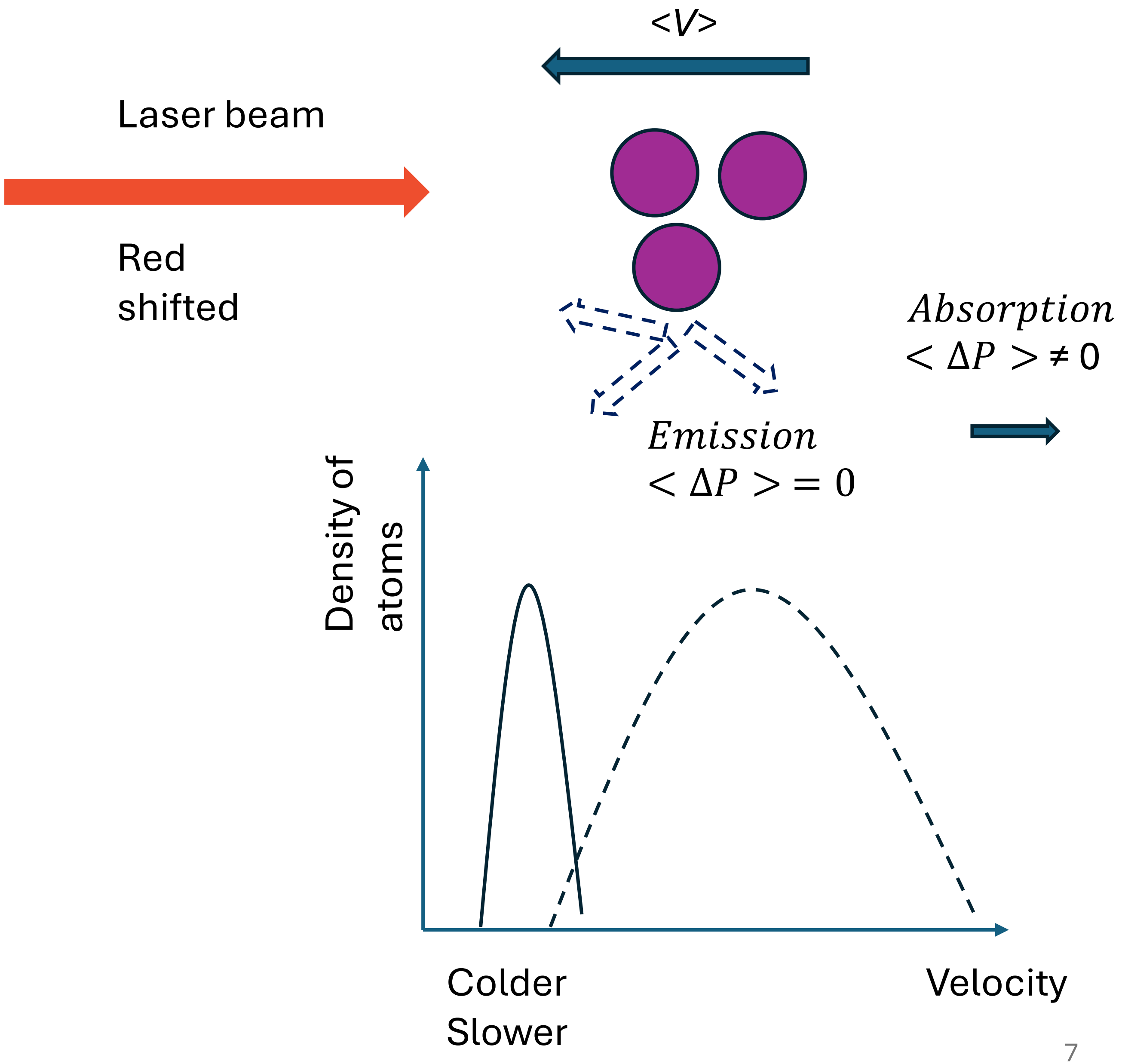

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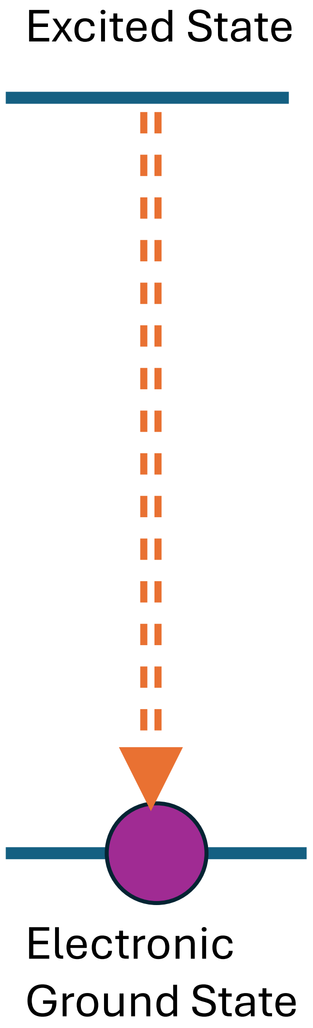
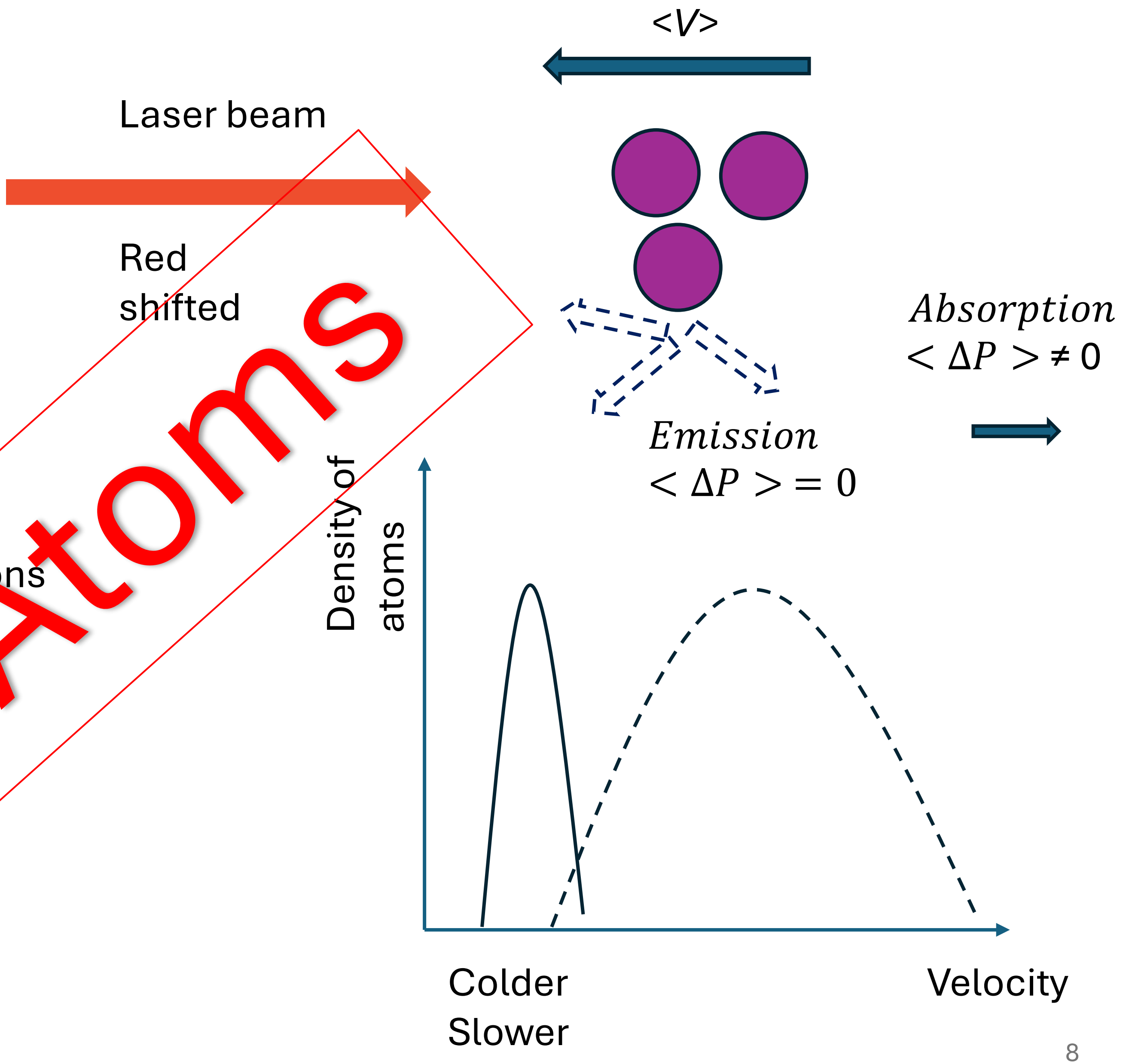
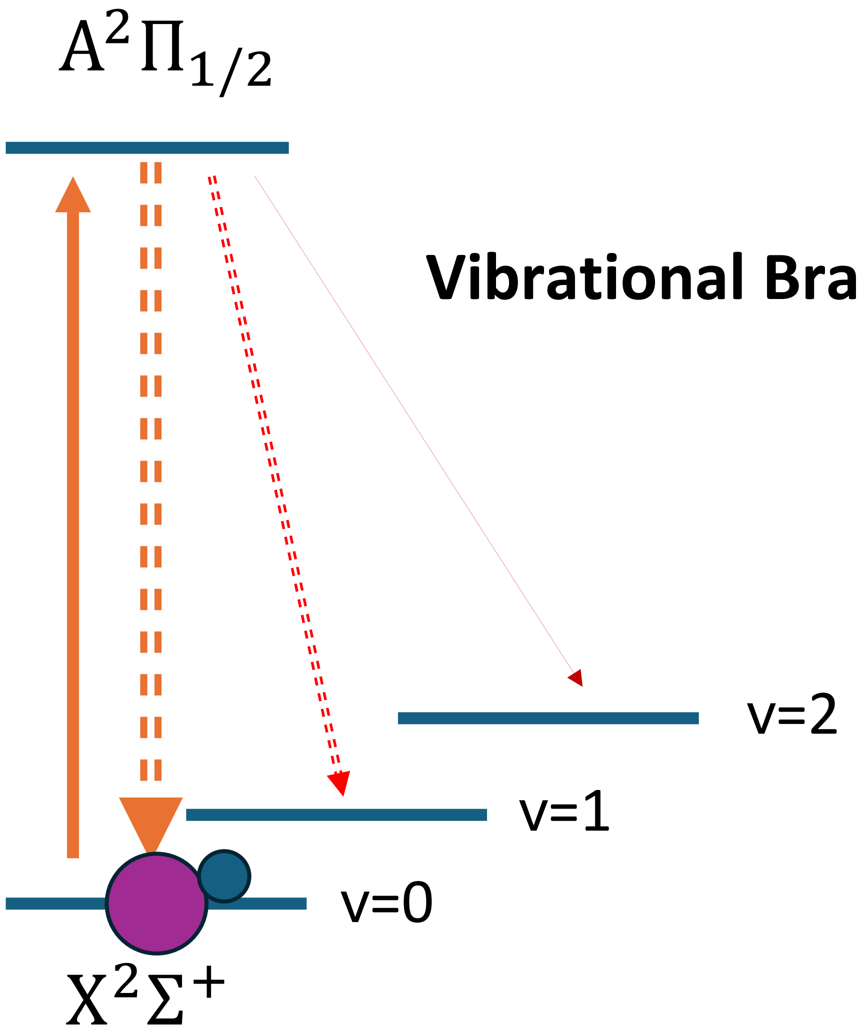


For reference, laser cool Rb
from 300 K to 1 mK
Need scatter ~ 30 000 photons
per atom

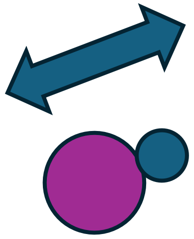
Atoms



Laser Cooling Molecules

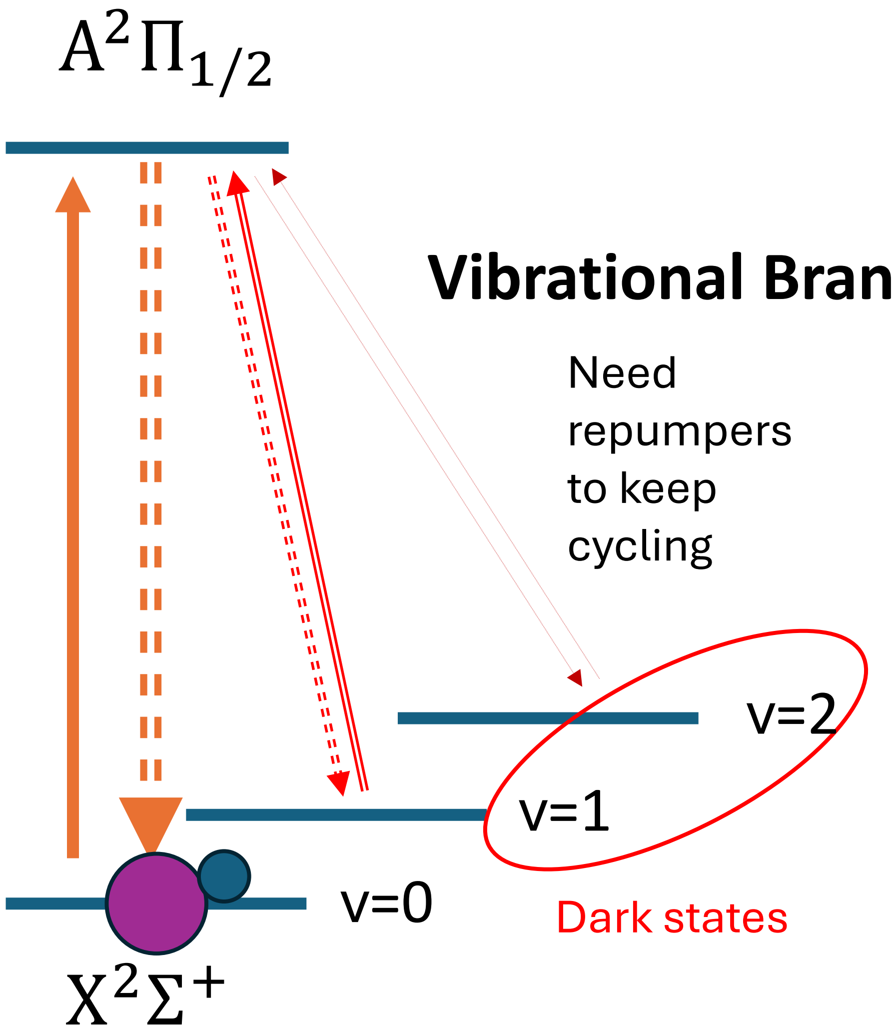


Vibrational Branching



Vibrational degree of freedom

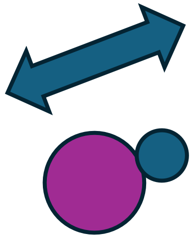
Laser Cooling Molecules



Vibrational Branching

Need repumpers to keep cycling

Luckily, can choose molecules that have electronic transitions with high probabilities of decaying back to same vibrational states



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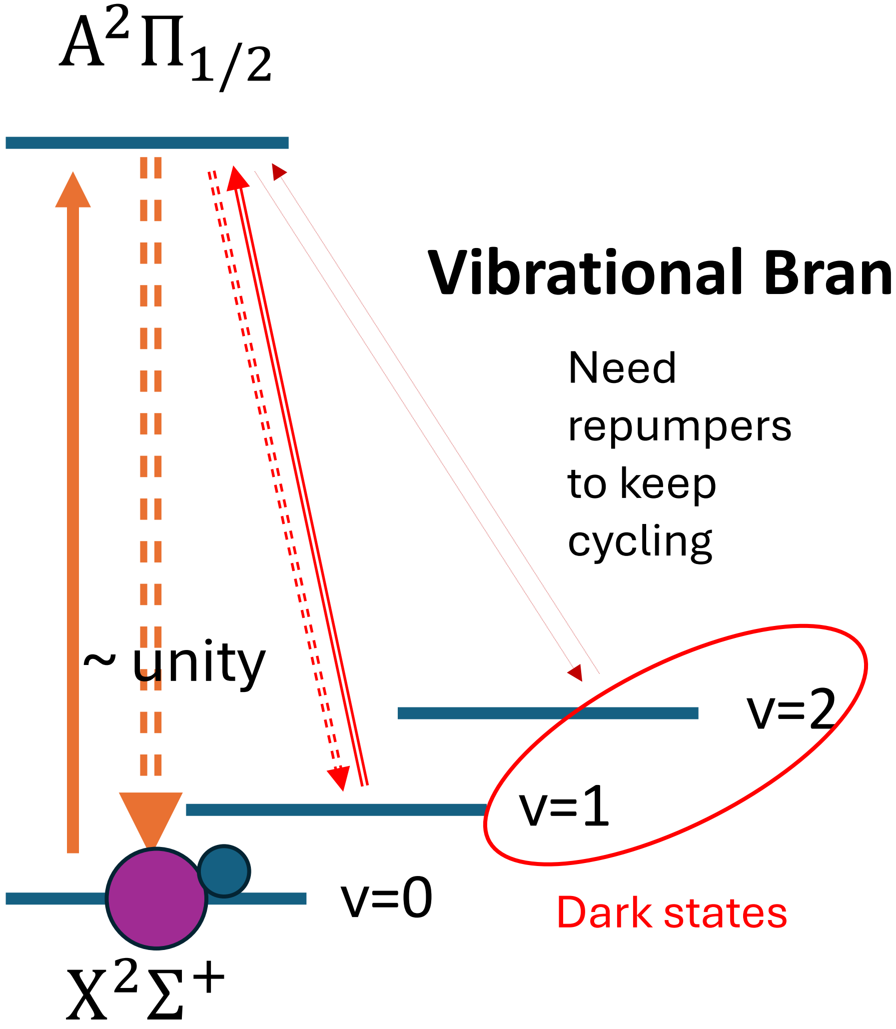

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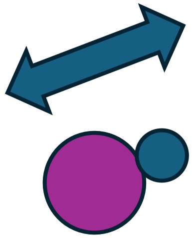

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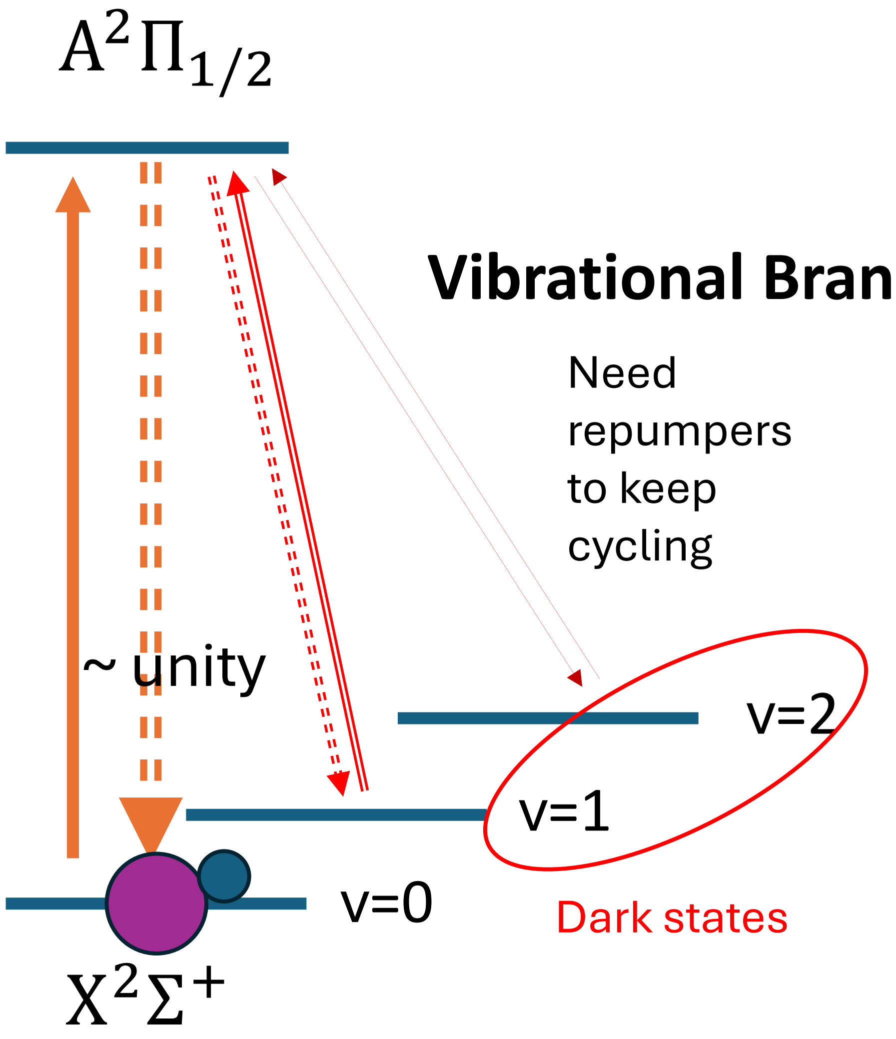
Laser Cooling Molecules



Vibrational Branching

Need repumpers to keep cycling

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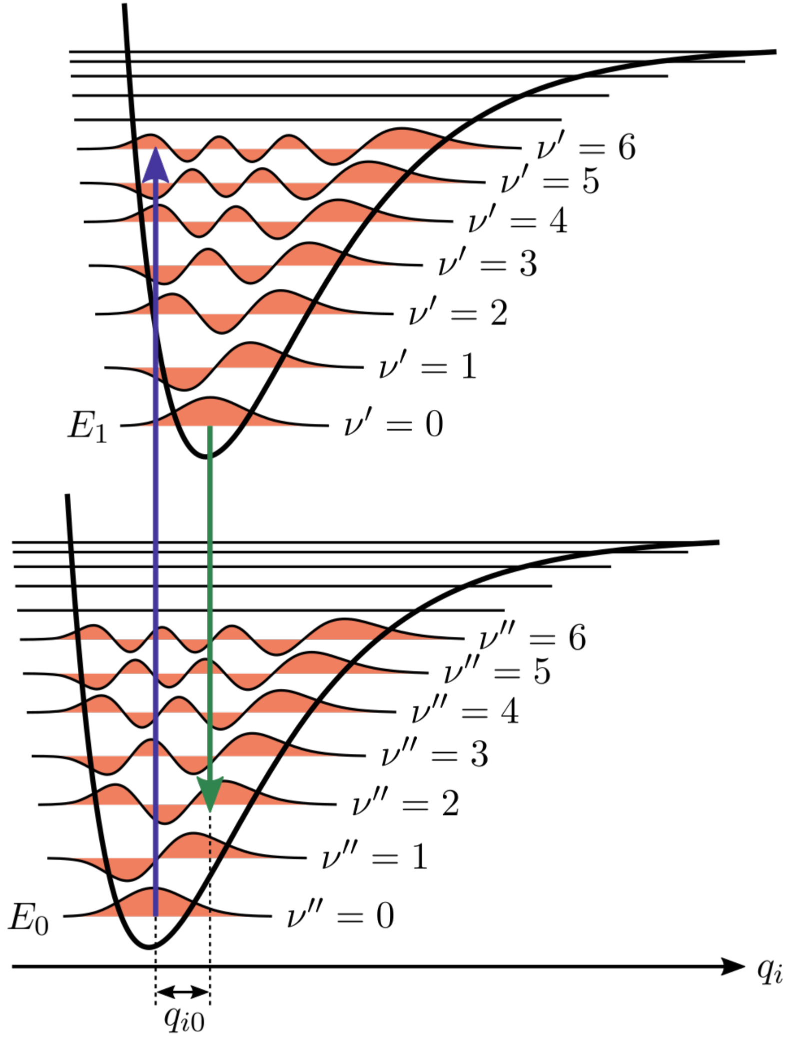
This usually happens when there is a heavy atom in polar molecules : eg, YO, CaF, SrF ...

Electron motions decoupled from vibration motions

Quantitative measure: Franck-Condon Factors (FCF)

FC factors

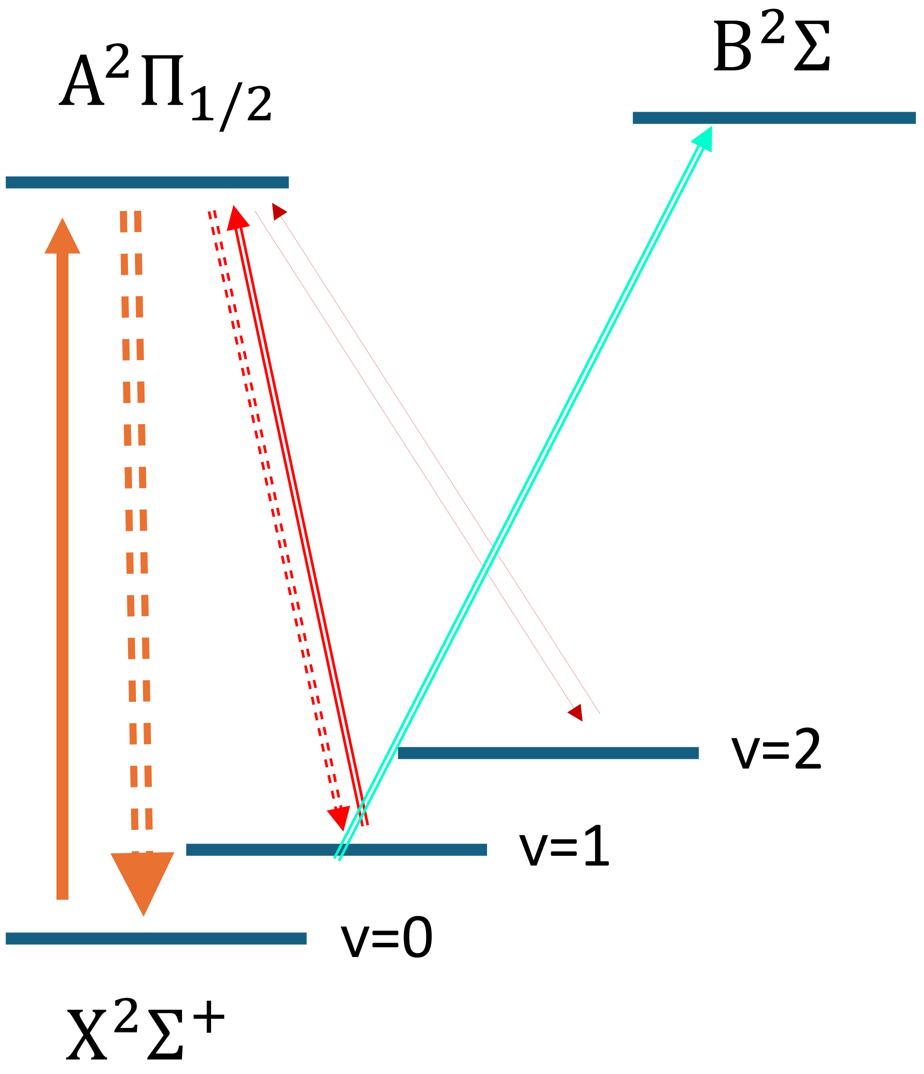


FCF :

Square of the overlap of the integral under the wavefunctions

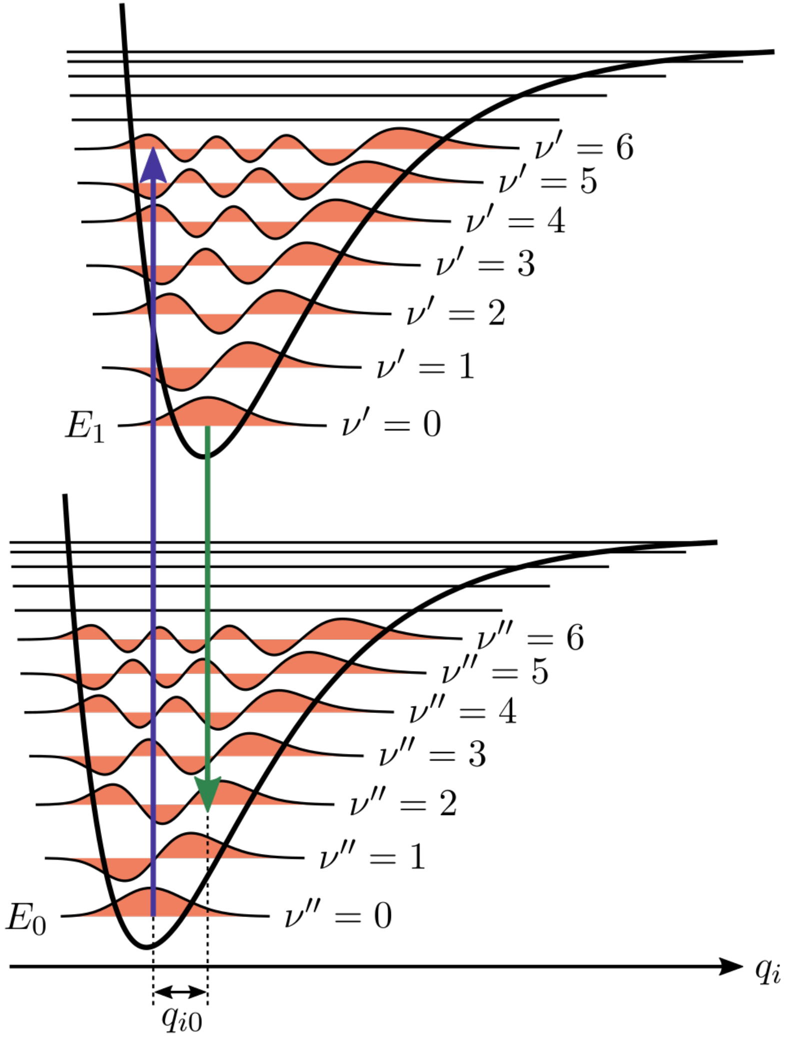
$$q_{v'v'} = |\langle \psi_{v'} | \psi_{v''} \rangle|^2,$$

Upgrading YO laser cooling scheme



Faster scattering rate
But need to make sure FCF are workable.

RKR inversion



The RKR (Rydberg–Klein–Rees) procedure (Rydberg, 1931;
Klien, 1932; Rees, 1947) : semiclassical procedure for determining
the two distances which correspond to the classical turning
points of the vibrational energy level.

Compare to other methods in quantum chemistry to calculate potentials:
Take high resolution spectroscopy values -> order of magnitude more accurate

Calculation Procedure

The RKR Inversion Method ©

by

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Calculation Procedure

- Take the vibrational energy and rotational constants extracted from the high-resolution spectroscopy papers of YO (A->X, B->X) transitions

TABLE 3

EQUILIBRIUM CONSTANTS (cm^{-1}) FOR THE X, A, AND B STATES OF YO

	$X^2\Sigma^+$	$A^2\Pi$	$B^2\Sigma^+$
T_e	0	16 529.75	20 793.33
ω_e	861.461	820.337	758.73[a]
$\omega_e x_e$ ...	2.867	3.377	3.97[a]
$\omega_e y_e$ ...	0.006_2	0.014_3	
B_e	0.38886_1	0.38678[b]	0.37314
α_e	0.00172_2	0.00203_1 [b]	0.00249
r_e(Å) ...	1.7880	1.7928[b]	1.8252

[a] Iteratively calculated values from Pekeris's formula.
[b] These values are from BBL.

A.Bernard and R.Grabivina 1980

Calculation Procedure

- Fit the vibrational energies and rotational constants (obtained from spectroscopy) to continuous functions G(v) and B(v)

G(v) is the vibration energy term

$$G(v) = \omega_e \left(v + \frac{1}{2}\right) - \omega_e x_e \left(v + \frac{1}{2}\right)^2 + \omega_e y_e \left(v + \frac{1}{2}\right)^3 + \cdots$$

B(v) is the rotational constant

$$B_v = B_e - \alpha_e \left(v + \frac{1}{2}\right) + \gamma_e \left(v + \frac{1}{2}\right)^2 + \cdots$$

Calculation Procedure

- From functions G(v) and B(v), can calculate the two classical turning point bond lengths, the inner r and outer r as function of v.
- The potential surface can then be calculated by varying v and interpolate the points. V(r(inner)) = V(r(outer)) = G(v)
- Once you have the potential surface, write shrodinger equation in matrix form solve it for v=0, v=1 ...

$$q_{v'v'} = |\langle \psi_{v'} | \psi_{v''} \rangle|^2,$$

Plan

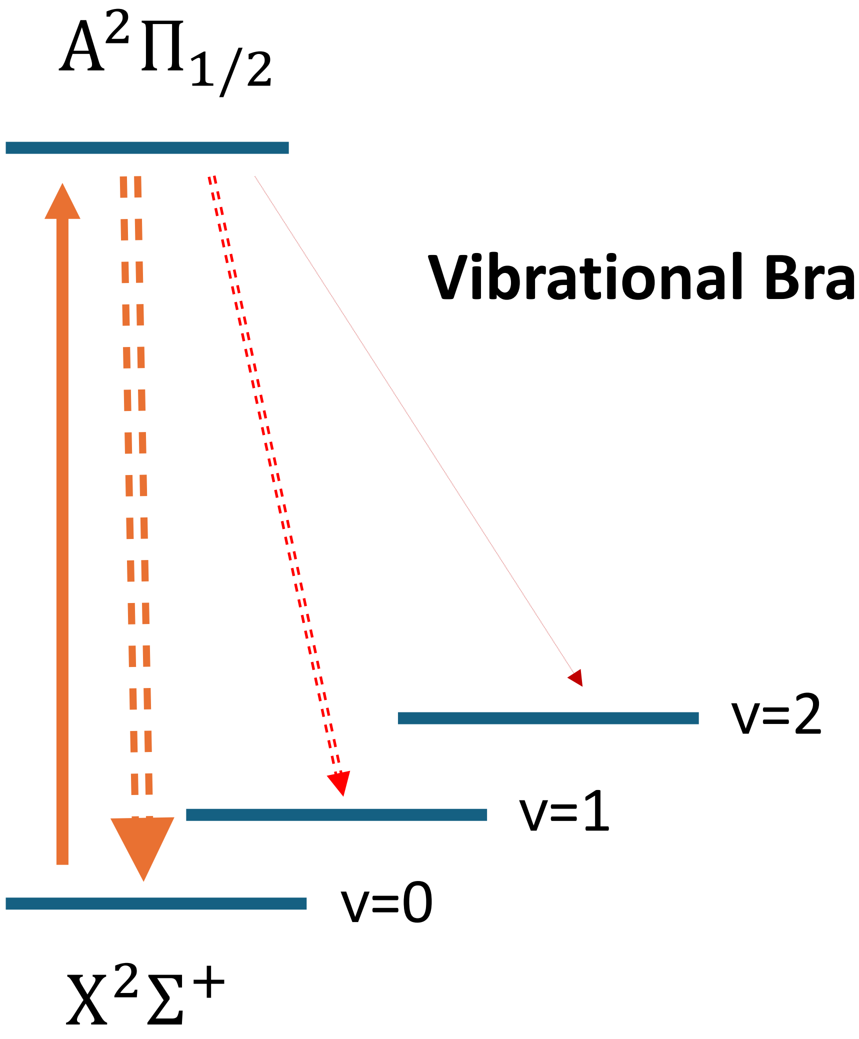
- Current status: able to calculate the FCF between B Sigma and X Sigma states
- Need experimental measurement to confirm

- Objective:

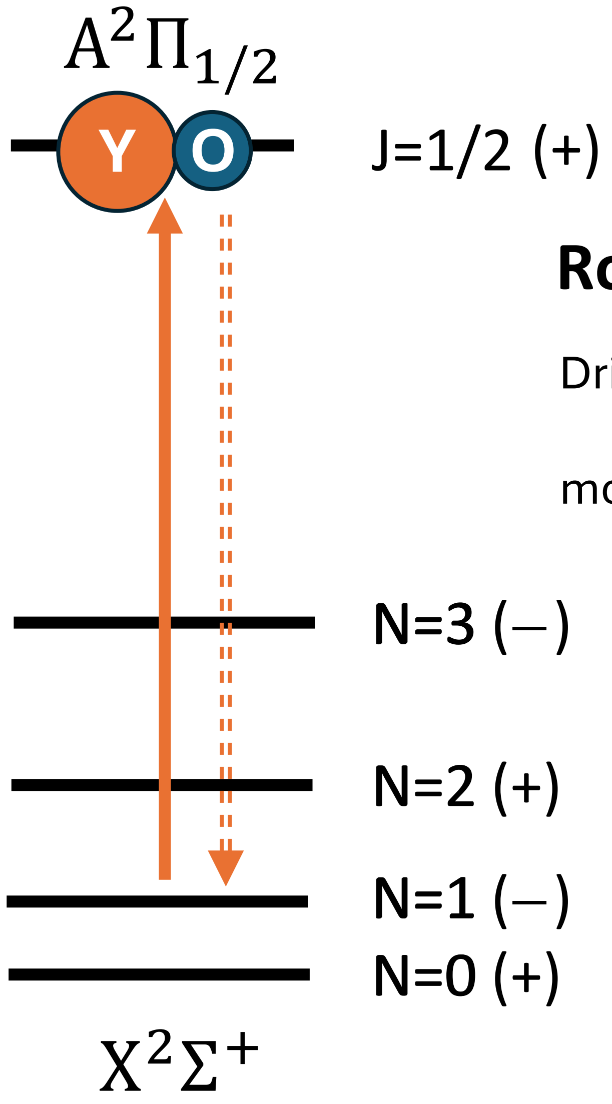
1. Figure out a better way to set bounds of wavefunctions
2. Calculate the uncertainties (different spectroscopy papers reports different constants with uncertainties)
3. Tighten up the code to make it more convenient to use for arbitrary transitions.

Thank you!

Molecular Laser Cooling



Vibrational Branching



Rotational Closure

Driving to an excited state with lower angular momentum is known as a “Type II Transition”

Laser Cooling of Atoms